Moving Object Extraction Using Background Separation: First Interim Report

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2026

1. Research Area and Motivation

Nowadays, the problem of separating moving objects from static background is becoming increasingly vital, because reliable scene understanding from cameras has become a core requirement in many critical systems. In military technology, moving object extraction is used for border monitoring, UAV-based target tracking, and early threat detection in complex environments. In civilian domains, the same methods are essential for smart-city surveillance, traffic analytics, autonomous and assisted driving, industrial safety monitoring, robotics, medical and sports video analysis, and modern video editing workflows. Improving robustness and efficiency of these methods therefore has both scientific value and immediate practical impact.

The topic relies on Linear Algebra concepts such as Low Rank Approximation, Singular Value Decomposition (SVD), Robust Principal Component Analysis (PCA). These tools allow us to model, analyze, and solve the problem in a mathematically structured way.

2. Aim of the Project and Main Tasks

The main aim of the project is to develop the algorithm with understandable user interface, which will allow to decompose input videos into two output videos: background (static part) and foreground (moving objects). To achieve this aim, the project is divided into the following tasks:

1. **Initial video processing** — converting input video file into data matrix $A := [f_1, \dots f_n]$, where each f_k corresponds to k^{th} frame of the video in grayscale. Moreover, $f_k \in \{0, 1, \dots 255\}^{w \cdot h}$, where w, h is the width and height of each frame in pixels

2. **Finding $\mathbf{L}, \mathbf{S} : \mathbf{A} = \mathbf{L} + \mathbf{S}$** — from the Low Rank Assumption, L is the low-rank part of A (in our case, background), and S is sparse matrix consisting of parts that vary a lot across frames.

3. **Post-processing** — displaying videos from found matrices L and S

4. **User interface development** — developing an intuitive user interface around the working algorithm.

3. Possible Approaches

There are several possible approaches to this problem.

3.1 Grassmannian Methods for RPCA [1]

Grassmannian RPCA methods perform low-rank estimation by optimizing directly on the Grassmann manifold, which represents the space of all possible subspaces of a given dimension.
Advantages: This approach is suitable for online and streaming data, including real-time video processing, and avoids full matrix decomposition.
Drawbacks: It requires prior knowledge or reliable estimation of the low-rank subspace rank, and optimization on manifolds is algorithmically complex.

3.2 Inexact Augmented Lagrange Multiplier Method (IALM) [2]

The Inexact Augmented Lagrange Multiplier method is an iterative algorithm designed to efficiently solve the optimization problem introduced by PCP. **Advantages:** This method is easy to implement in code and demonstrates good convergence properties in practical RPCA problems.
Drawbacks: Performance depends on appropriate selection of penalty parameters, and it is not ideal for very large or streaming datasets without modification.

3.3 Chosen Solution

Based on this review, our team is planning to solve the background subtraction problem using the approach described in **3.2**. This method should be easy to implement in code, because each step of the algorithm could be nicely translated into code and it is not heavy theoretically. The main challenge in the implementation stage would be finding the correct Lagrange Multiplier as a threshold, which will be done after precise fine-tuning and testing on video streams.

4. Current Progress

So far, we have already recorded and submitted the initial workshop videos, describing the main steps of the algorithm.

5. Pseudocode of the Implementation

Below is the high-level pseudocode of the planned implementation.

```
Input: Video file (.mp4, .mov, etc.)
Output: Two video files:
background and moving objects videos

1. Preprocess the data & construct the matrix
representation A
2. Solve the decomposition problem: A = L + S
via Inexact Augmenthed Lagrange Multiplier
method
3. Refine the foreground component S
4. Evaluate the detection quality
5. Display the results
```

6. Testing Plan and Data Availability

To test the implementation, we will use Background Models Challenge dataset and our own dataset of videos recorded on campus. BMC dataset consists of videos for testing our background subtraction algorithm. BMC propose two datasets: Learninig mode, with synthetic videos, and Evaluation mode, with complex synthetic and real videos. Our own dataset is at the stage of collecting tha data.

The evaluation will be based on second dataset:

- **Correctness:** BMC provides "ground truth" (binary masks) for many of its videos, which allows us to calculate the error rate of our algebraic model by comparing our sparse matrix S to the actual moving objects
- **Performance:** We will report average processing time per frame (ms/frame), total runtime per video, and peak memory usage. We will also analyze scalability by testing videos with different resolutions and frame counts.
- **Quality of results:** Using BMC ground-truth masks, we will measure precision, recall, F1-score, and Intersection over Union (IoU) for foreground detection. Additionally, we will report reconstruction error via Frobenius norm $\|A-(L+S)\|_F/\|A\|_F$ and include visual comparisons of extracted foreground masks.

7. Future Research Plan and Time Schedule

The next steps of the project are:

Time period	Planned work
Week 1	Collection of remaining data
Week 2	Matrix Construction & Data Preprocessing
Week 3–4	Implementing algorithm
Week 5	Testing & Debugging
Week 6	Presenting

8. Possible Challenges

There are several challenges that might hamper the work. One of them could be data-related challenges(especially real data), e.g. camera jitters, illumination changing, sudden shadows on a video, resolution artifacts. Also there could be difficulties in computational processing.

9. Conclusion

At this stage, the project is focused on collecting the data and researching the topic and algorithm. The main progress so far includes deep basic understanding of the project idea and ways of implementing it. The next phase will concentrate on actual implementing the code for project and testing the results.

Overall, the project is expected to demonstrate how linear algebra methods can be used to detecting moving objects in video streaming using background separation.

References

[1] Balzano, L., Nowak, R., and Recht, B. (2010). *Online Identification and Tracking of Subspaces from Highly Incomplete Information.* Proceedings of the 48th Annual Allerton Conference on Communication, Control, and Computing, Monticello, IL, USA, pp. 704–711. `https://www.emergentmind.com/papers/1006.4046`

[2] Xin, B., Tian, Y., Wang, Y., and Gao, W. (2015). *Background Subtraction via Generalized Fused Lasso Foreground Modeling.* Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4676–4684. `https://openaccess.thecvf.com/content_cvpr_2015/papers/Xin_Background_Subtraction_via_2015_CVPR_paper.pdf`